Revision Session – Supervised and Unsupervised Learning Algorithms

Use the following dataset representing 10 students' performance in a course. Features include:

- Study Hours (weekly average hours studied)
- Attendance Percentage (percentage of classes attended)
- Previous Exam Score (out of 100)
- Gender (Male or Female)
- Extracurricular (Yes or No, indicating participation in activities)

Student ID	Study Hours	Attendance Percentage	Previous Exam Score	Gender	Extra-Curricular Activities	Result
1	5	80	65	Male	No	Pass
2	3	60	50	Female	No	Fail
3	6	70	70	Female	Yes	Pass
4	1	50	40	Male	Yes	Fail
5	4	75	60	Male	Yes	Fail
6	8	90	85	Female	Yes	Fail
7	7	85	80	Male	No	Pass
8	9	95	90	Female	Yes	Pass
9	2	80	50	Male	Yes	Fail
10	10	98	95	Male	Yes	Pass

a) Identify which columns denote categorical and numerical features.
Categorical: Gender, Extra-Curricular Activities.
Numerical: Study Hours, Attendance Percentage, Previous Exam Score.

b) Calculate the Gini index for "Gender" and "Extra-curricular Activities" feature columns. Show the steps in your calculations.
For Gender:

Males (6 samples, 3 Pass, 3 Fail)

Gini = 1 - (3/6)^2 - (3/6)^2 = 0.5.

Females (4 samples, 2 Pass, 2 Fail)

Gini = 0.5.

Weighted Gini = (6/10)*0.5 + (4/10)*0.5 = 0.5.

For Extra-Curricular Activities:

No (3 samples, 2 Pass, 1 Fail),

Gini = 1 - (2/3)^2 - (1/3)^2 = 0.44.

Yes (7 samples, 3 Pass, 4 Fail),

Gini = 1 - (3/7)^2 - (4/7)^2 = 0.49.

Weighted Gini = (3/10)*0.44 + (7/10)*0.49 = 0.48.

c) Based on part (b) result, which feature is most suited to take as the root node in generating the decision tree? Justify your answer.
Extra-Curricular Activities (lower Gini 0.48 < 0.5, better purity after split)

d) The above dataset was split into training and testing sets, with the first 5 rows used for training and the remaining rows for testing. Based on a classifier trained on this split, the predicted values for the test set (Students 6 to 10) are provided.

Student ID	Actual Result	Predicted Result
6	Fail	Pass
7	Pass	Fail
8	Pass	Pass
9	Fail	Fail
10	Pass	Fail

i. Create the confusion matrix.

	Pred Pass	Pred Fail
Act Pass	1	2
Act Fail	1	1

ii. Calculate the Accuracy of the classifier.
Accuracy = (1+1)/5 = 0.40

iii. Interpret what each metric Accuracy, Precision, Recall and F1-Score indicate about the classifier's performance in predicting student passing.

- Accuracy (0.40): % correct predictions.
- Precision (0.50): % predicted Pass that are true.
- Recall (0.33): % true Pass captured.
- F1 (0.40): balance of precision/recall.
- Classifier has low accuracy, moderate precision, low recall. Not a good classifier.

e) Calculate the means and standard deviations for each numerical column you identified in part (a).

Column	Mean	STD
Study Hours	5.50	2.87
Attendance	78.30	14.39
Prev Score	68.50	17.76

f) Create the scaled dataset using Standardization method only for the numerical columns. ($Z = \frac{x-\mu}{\sigma}$)

Student ID	Study Hours	Attendance Percentage	Previous Exam Score
1	-0.17	0.12	-0.20
2	-0.87	-1.27	-1.04
3	0.17	-0.58	0.08
4	-1.57	-1.97	-1.60

Student ID	Study Hours	Attendance Percentage	Previous Exam Score
5	-0.52	-0.23	-0.48
6	0.87	0.81	0.93
7	0.52	0.47	0.65
8	1.22	1.16	1.21
9	-1.22	0.12	-1.04
10	1.57	1.37	1.49

g) Compute the Euclidean distances between Student 1 and Student 6 and Student 1 and Student 9 using the standardized dataset. Show the steps in your calculations.

Dist 1-6:

$\sqrt{[(-0.17-0.87)^2 + (0.12-0.81)^2 + (-0.20-0.93)^2]}$

$= \sqrt{[1.08 + 0.48 + 1.28]} = \sqrt{2.84} = 1.68$

Dist 1-9:

$\sqrt{[(-0.17-(-1.22))^2 + (0.12-0.12)^2 + (-0.20-(-1.04))^2]}$

$= \sqrt{[1.10 + 0 + 0.71]} = \sqrt{1.81} = 1.34.$

h) Based on the results from part (g), identify which student is the most suitable to merge with Student 1 into the same cluster. Justify your answer.

Student 9 (smaller dist 1.34 < 1.68) Smaller distance implies clusters are closer to one another. Student 9 is closest to student 1. We can merge them together

i) For k-means clustering (k=2) on the standardized dataset obtained in part(f), the clusters and their centroids are given below.

Cluster	Student ID	Centroid Coordinates
Cluster 1 – C1	1, 2, 3, 4, 5	(-0.871, -0.785, -0.867)
Cluster 2 – C2	6, 7, 8, 9, 10	(0.871, 0.785, 0.867)

Answer the following:

i. Compute the inertia (within-cluster sum of squares) for the two clusters separately using the equation given below. Show calculations for each cluster and sum them to get the total inertia.

$$Inertia = \sum_{points\ in\ cluster}(Euclidean\ distance\ to\ centroid)^2$$

Cluster	Student ID	Euclidean Distance to Centroid	Squared Distance
C1(-0.871, -0.785, -0.867)	1	1.32	1.75
C1	2	0.51	0.27
C1	3	1.42	2.02
C1	4	1.56	2.43
C1	5	0.76	0.58
C1 Total			**7.05**
C2(0.871, 0.785, 0.867)	6	0.02	0.00
C2	7	0.52	0.27
C2	8	0.62	0.38
C2	9	2.91	8.45
C2	10	1.11	1.22
C2 Total			**10.32**
Total Inertia (C1+C2)			**17.37**

ii. Compute the Dunn index for the above clustering. Show all calculations.

Max intra C1: 1.56

Max intra C2: 2.91

Max intra-cluster distance overall: **2.91**

Min inter-cluster distance=

$$\sqrt{[(0.871 - (-0.871))^2 + (0.785 - (-0.785))^2 + (0.867 - (-0.867))^2]}$$

$$= \sqrt{[(1.742)^2 + (1.570)^2 + (1.734)^2]}$$

$$= \sqrt{[3.0346 + 2.4649 + 3.0068]} = \sqrt{8.5063} \approx \mathbf{2.92}$$

Dunn = 2.92 / 2.91 ≈ 1.00

iii. Explain what the calculated Dunn index value indicates about the quality of the clusters.

The Dunn Index of 1.00 indicates moderate cluster quality. The clusters are reasonably separated (inter-cluster distance 2.92 ≈ maximum intra-cluster distance 2.91), but not highly compact, suggesting some overlap or spread, particularly in Cluster 2. The clustering is acceptable but could be improved for better separation and compactness.